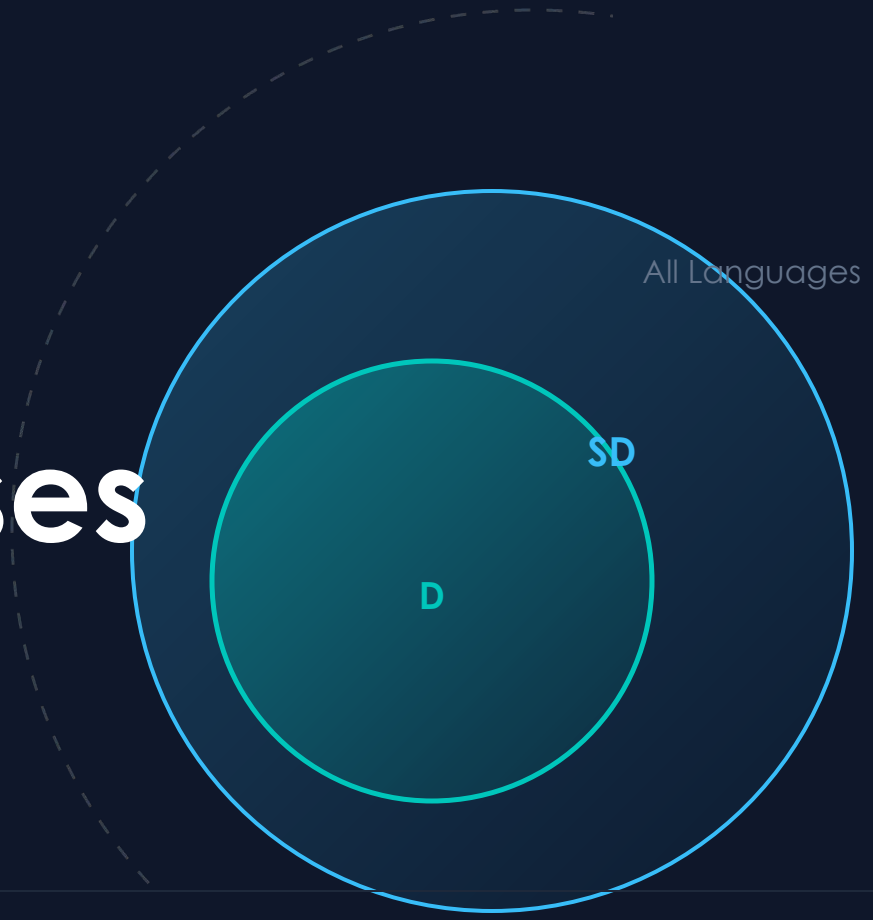
Theoretical CS

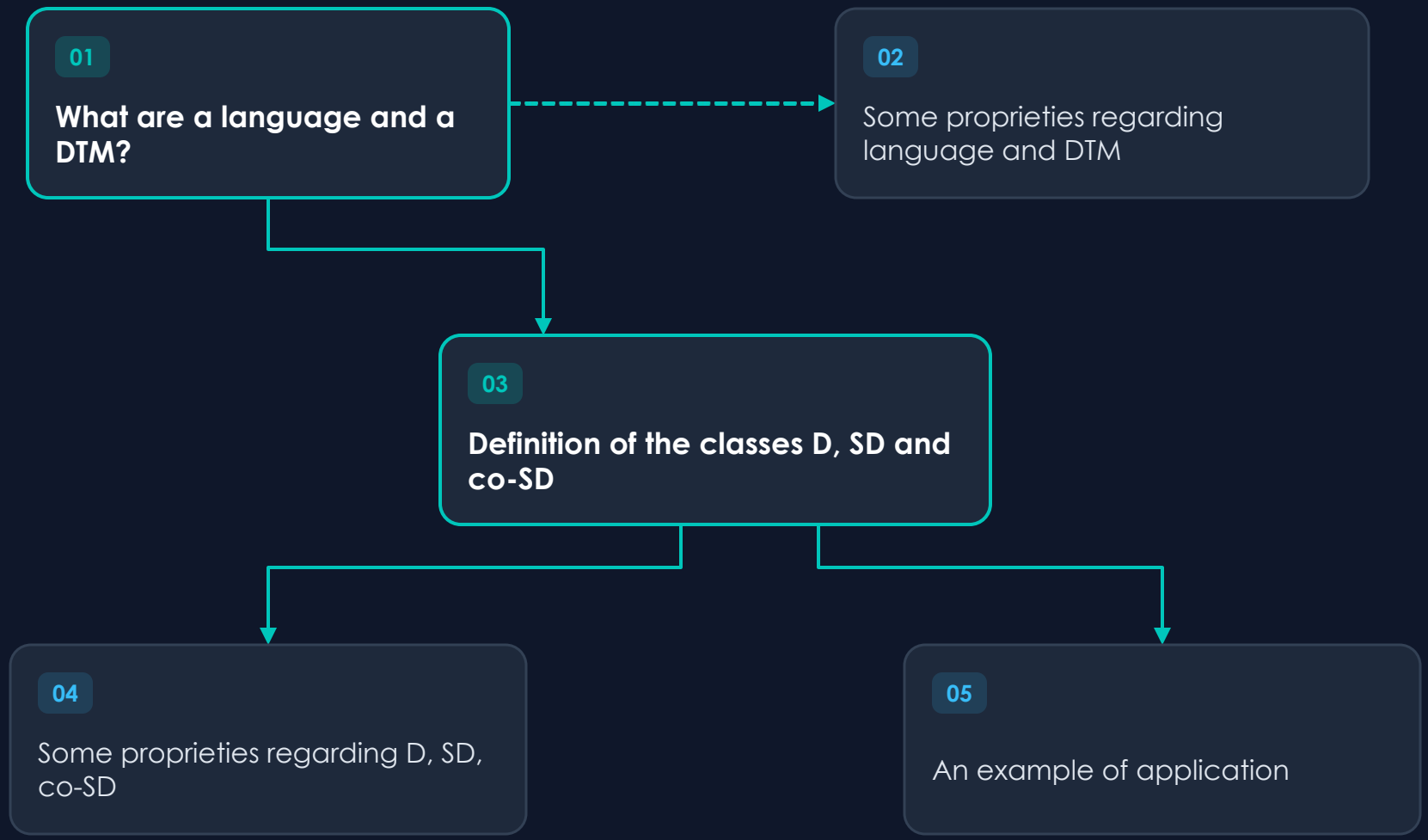
Decidability classes

A formal classification of language complexity & computability



STRUCTURED FLOW

Conceptual scheme



THEORETICAL CS

What is a language?

A language, from an intuitive point of view, is a **set of strings** composed of symbols belonging to a formal alphabet. It bridges abstract computation with structured representation.

01

Binary Language

Built on the alphabet of symbols **{0, 1}**. The foundation of digital systems and computer architecture.

02

Natural Language Lexicon

Built on standard alphabets like **{a, b, c, ..., z}**. Represents structured human language dictionaries.

FORMAL FOUNDATIONAL DEFINITIONS

Definition: Alphabet (Σ)

An alphabet is a **totally ordered set** of **finite cardinality**. It serves as the atomic building blocks of a language system.

Definition: String (w)

A string is a **finite sequence** of elements chosen from an arbitrary alphabet. Its length is always non-negative.

THEORETICAL CS

Deterministic Turing Machines (DTM)

A deterministic Turing machine is a mathematical formalism that allows us to simulate computer behavior, formally defined as a 6-tuple: **(K, Σ, Γ, δ, H, s)**

K — States Set
Finite set representing all possible internal states of the machine.

Σ — Input Alphabet
Alphabet of input symbols. It strictly does not contain the blank symbol (□).

Γ — Tape Alphabet
Contains □, the input alphabet Σ, and potentially other tape symbols.

s — Initial State
The starting state of the machine, where $s \in K$.

H — Halting States
Subset of states ($H \subseteq K$) that cause execution to stop.

δ — Transition Function
$\delta: (K \setminus H) \times \Gamma \rightarrow K \times \Gamma \times \{\leftarrow, \rightarrow\}$
Maps current non-halting state and tape symbol to next state, write symbol, and direction.

SPECIALIZED FUNCTIONAL CLASSIFICATIONS

Decider Machine
A machine M is defined as a decider if its set of halting states consists strictly of acceptance and rejection: $H = \{y, n\}$.

Transducer Machine
A machine M is defined as a transducer if it computes a function and halts in a single designated state: $H = \{h\}$.

THEORETICAL CS

The Power of Encoding

01

Encoding of an Alphabet

Each alphabet **Σ** can be systematically encoded into binary representation.

- Symbols of Σ are mapped to strings over **{0,1}***
- Binary string length is strictly bounded by **⌈log₂ | Σ | ⌉**
- Establishes a uniform digital baseline for language symbols

02

Encoding of a DTM

Every Deterministic Turing Machine can be represented entirely as a string.

- Encodes machine states, alphabets, and transition function **δ**
- Serializes full machine structure into a single string
- As a result, the set of all DTMs constitutes a formal **language**

03

Cardinality of Languages

Any formal language can be mapped as a sub-language of **{0,1}***.

- Since **{0,1}*** is countably infinite
- Every language possesses countable or finite cardinality
- Implies the set of all possible DTMs is also **countable**

THEORETICAL CS

Language of a DTM

Intuitively, for any input string **w**, a DTM-decider **M** exhibits one of three distinct behaviors:

M(w) = y

Accepts

Takes input **w** and reaches the halting state **'yes'** in finite time.

M(w) = n

Rejects

Takes input **w** and reaches the halting state **'no'** in finite time.

M(w) = ↑

Loops

Does **not accept or reject w**, running indefinitely without halting.

FORMAL LANGUAGE DEFINITION

From these behaviors, we define the formal language **L(M)** accepted by a DTM **M** as:

$$L(M) = \{ w \in \Sigma^* \mid M(w) = y \}$$

THEORETICAL CS

The classes D, SD, co-SD

From the concept of acceptance and rejectance, it is natural to introduce the following three classes of D, SD, and co-SD languages:

Class D

Decidable Languages

D is the class of languages **L**, for which there exists a DTM **M** such that:

- **M accepts** all strings of **L**
- **M rejects** all strings of **¬L**
- Always halts in finite time for any input

Class SD

Semi-Decidable Languages

SD is the class of languages **L**, for which there exists a DTM **M** such that:

- **M accepts** all strings of **L**
- May reject or loop infinitely on strings not in **L**
- Guaranteed halting only on positive instances

Class co-SD

Co-Semi-Decidable Languages

co-SD is the class of languages **L**, for which there exists a DTM **M** such that:

- **M rejects** all strings of **¬L**
- May accept or loop infinitely on strings in **L**
- Guaranteed halting only on negative instances

THEORETICAL CS

Good Questions

After introducing these three classes, you can ask yourself a few questions such as:

QUESTION 01

What are the relationships between D, SD and co-SD?

QUESTION 02

How ‘big’ are they?

QUESTION 03

Does an arbitrary language always belong to one of these classes?

QUESTION 04

What is their potential use outside of theoretical computing?

In the next slides we will give an answer to all these questions.

THEORETICAL CS

Some relationships between D, SD and co-SD

It is evident from the definitions of D, SD, and co-SD that these three classes are closely related, in fact we can represent via Venn diagrams as follows:



THEORETICAL PROOF

Equivalence: D = SD ∩ co-SD

Because **D = SD ∩ co-SD**, it is evident from the definition that **D ⊆ SD ∩ co-SD**.

The inverse inclusion is verified because for each **L ∈ SD ∩ co-SD** via **M_1** (machine that semi-decides L) and **M_2** (machine that co-semi-decides L), we can build **M_3** that decides L.

ALGORITHM

Decision Procedure

```
Boolean decide_SD ∩ co-SD(M₁, M₂, w) {
  for i in ℕ
    if(utms(M₁, w, i) == y) return true
    if(utms(M₂, w, i) == n) return false
  end
}
```

THEORETICAL CS

Relationships Between D, SD, and co-SD

Analysis of non-emptiness properties and structural collapse under the Halting Problem hypothesis.

HALTING PROBLEM

Non-emptiness: SD \ D and co-SD \ D

- The Halting Problem **H** is semi-decidable but not decidable (**H ∈ SD \ D**).
- Consequently, **¬H** belongs to **co-SD \ D**.

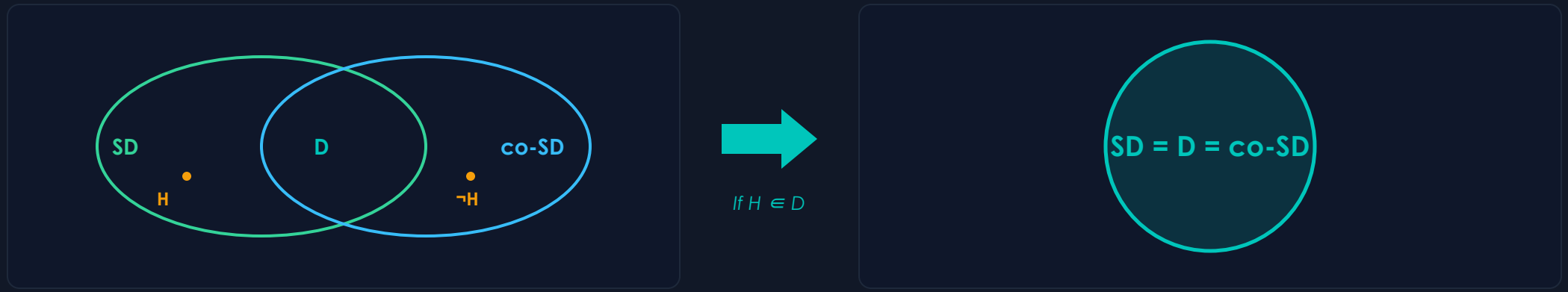
Note: For all languages L, if L ∈ SD then ¬L ∈ co-SD.

COLLAPSE HYPOTHESIS

If H ∈ D, then SD = D = co-SD

- If by contradiction **H ∈ D**, then SD and co-SD would collapse into D.
- This is because we could determine in finite time whether a DTM on a string loops or not and act accordingly.

VISUAL REPRESENTATION



THEORETICAL CS

The Cardinality of D, SD, and co-SD

Mathematical proofs establishing that D, SD, and co-SD are all countably infinite sets.

PROPOSITION

SD is Countable

Proof: Let **M** be the set of all Deterministic Turing Machines (DTMs).

As established previously, the set **M** is countable.

Consider the mapping function:

$$f : M \to SD$$

where each DTM M associates to L(M)

By definition, every language in SD is semi-decided by at least one DTM, so **f** is a **surjective function**.

This implies the cardinality constraint:

$$|SD| \leq |M|$$

Since **M** is countable, it follows directly that **SD is countable**.

COROLLARY

D and co-SD are Countable

Proof for D:

D is a non-finite proper subset of **SD** (**D ⊆ SD**). Since any subset of a countable set is countable, **D is countable**.

Proof for co-SD:

We construct an explicit bijection from **SD** to **co-SD**:

$$f : SD \to co\text{-}SD$$

such that f(L) = ¬L

Since complementation is a bijection, this directly implies equal cardinality:

$$|co\text{-}SD| = |SD|$$

Since **SD** is countable, **co-SD is also countable**.

THEORETICAL CS

Does Every Language Belong to SD or co-SD?

Cardinality proof establishing that unrecognizable languages vastly outnumber semi-decidable ones.

ALL LANGUAGES

Power Set $\mathcal{P}(\{0,1\}^*)$

Every formal language is defined as a subset of strings over binary alphabet:

$$L \subseteq \{0,1\}^*$$

The set of all possible languages is the Power Set:

$$\mathcal{P}(\{0,1\}^*)$$

By Cantor's Theorem, since {0,1}* is countably infinite, its power set is **uncountably infinite**:

$$|\mathcal{P}(\{0,1\}^*)| = 2^{\aleph_0} > \aleph_0$$

DECIDABILITY BOUNDS

Countability of SD ∪ co-SD

Both semi-decidable classes are countably infinite sets:

$$|SD| = \aleph_0 \text{ and } |co\text{-}SD| = \aleph_0$$

The union of any two countable sets remains **countably infinite**:

$$|SD \cup co\text{-}SD| \leq \aleph_0$$

Hence, the set of all machine-recognizable or co-recognizable languages is strictly bounded in size.

CONTRADICTION

Uncountably Many Outside

If every language belonged to SD or co-SD, then:

$$SD \cup co\text{-}SD = \mathcal{P}(\{0,1\}^*)$$

This presents an immediate contradiction: a countable set cannot equal an uncountable set!

Conclusion: The set difference is uncountable:

$$\mathcal{P}(\{0,1\}^*) \setminus (SD \cup co\text{-}SD)$$

"**Almost all**" formal languages belong neither to SD nor co-SD.

THEORETICAL CS

Gödel's First Incompleteness Theorem

Application of decidability concepts to formal mathematical logic and Peano arithmetic.

APPLICATION

Practical Relevance

Decidability concepts are fundamental across multiple areas of theoretical study.

In particular, they serve as a crucial basis for proving **Gödel's first incompleteness theorem**.

PEANO ARITHMETIC

Systemic Incompleteness

The theorem demonstrates that **Peano's axioms** are insufficient to prove all true propositions of number theory.

Even if additional axioms are added to Peano's system, it will **never be enough** to achieve complete provability.

PROOF MECHANISM

Decidability Disparity

- **Provable Propositions:** The set of all provable propositions is **semi-decidable (SD)**.
- **All Propositions:** The set of all propositions is **not semi-decidable**.

Conclusion:

Since one set is semi-decidable and the other is not, they **cannot be the same set**.

Thanks for the attention